

Paper 1

Part A

1. Role of MANETs in disaster recovery scenarios

MANETs (Mobile Ad-hoc Networks) play a crucial role in disaster recovery scenarios by providing temporary communication infrastructure, enabling rescue teams to coordinate and communicate effectively.

2. Primary function of sequence numbers in DSDV protocol

Sequence numbers in DSDV ensure the consistency of routing information by preventing outdated routing updates and resolving routing loops.

3. Contribution of reverse path forwarding in TBRPF protocol

Reverse path forwarding in TBRPF reduces routing update overhead by only sending updates along the reverse path, minimizing unnecessary transmissions.

4. Architecture of Wireless Mesh Networks

Not feasible to draw here, but it consists of mesh routers, mesh clients, and gateways, forming a dynamic, self-healing network.

5. Challenges of Wireless Mesh Networks

Scalability, interference, security, quality of service, and node mobility.

6. Definitions

- Multicasting: One-to-many communication, where a single sender transmits data to multiple receivers.

- Geo-casting: Location-based multicasting, where data is transmitted to all nodes within a specific geographic area.

Part B

1. Impact of Frequent Topology Changes on Proactive Routing Protocol (DSDV)

Frequent topology changes in a MANET with 50 nodes using DSDV will lead to:

- Increased control message overhead: Each topology change triggers the exchange of updated routing tables, resulting in higher control message overhead.

- Routing table inconsistencies: The frequent updates may lead to temporary inconsistencies in the routing tables, potentially causing routing loops or packet loss.

- Network congestion: The increased control message overhead and routing table updates can contribute to network congestion, affecting the overall network performance.

2. Route Discovery and Maintenance in Reactive Routing Protocol (AODV)

In AODV, the route discovery process for a 5-hop route takes:

- 5 seconds (1 second per hop)

If the route fails after 3 minutes, the process to discover a new route involves:

- Route error message propagation: The node detecting the route failure sends a route error message to the source node.
- Route request retransmission: The source node retransmits the route request to discover a new route.
- Impact of frequent route failures: Frequent route failures can lead to increased control message overhead, network congestion, and higher latency, affecting the overall network performance.

3. Working of Destination-Sequenced Distance-Vector (DSDV) Routing Protocol

DSDV is a proactive routing protocol that maintains a routing table with the shortest distance to each destination node. Here's an example:

Suppose we have a MANET with nodes A, B, C, and D. The routing table at node A is:

Destination	Next Hop	Distance
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B	B	1
C	B	2
D	C	3

When node A wants to send data to node D, it uses the routing table to determine the next hop (node C) and the distance (3 hops).

4. Ad Hoc Broadcast Protocol

Ad Hoc Broadcast Protocol is a protocol designed for broadcasting in MANETs. It aims to reduce the number of retransmissions and minimize the broadcast storm problem. The protocol typically involves:

- Controlled retransmissions: Nodes retransmit the broadcast message only if necessary.
- Neighbor knowledge: Nodes maintain information about their neighbors to optimize retransmissions.
- Adaptive transmission: The protocol adjusts the transmission strategy based on network conditions.

Part C

5. Greedy Forwarding in Position-Based Routing

In greedy forwarding, each node forwards the packet to the neighbor closest to the destination. Using the given node positions:

Node A (0, 0) -> Node B (3, 4) -> Node C (6, 8) -> Node D (10, 10)

The path is determined by each node selecting the neighbor closest to the destination (Node D).

6. Location Service in MANETs

To discover the location of node B, node A can use a Location Service, such as:

- Flooding-based location service: Node A floods the network with a location request, which is responded to by node B.
- Distributed location service: Node A queries a distributed location database to obtain node B's location.

Challenges in highly dynamic networks include:

- Location updates: Frequent node mobility requires frequent location updates.
- Location accuracy: Location information may become outdated or inaccurate due to node mobility.

7. Multicast Ad Hoc On-Demand Distance Vector (MAODV) Protocol

MAODV is a multicast routing protocol for MANETs. It handles route discovery and maintenance by:

- Creating a multicast tree: MAODV establishes a multicast tree connecting the source node to all group members.
- Maintaining the multicast tree: MAODV updates the multicast tree in response to node mobility and topology changes.

Challenges in highly dynamic networks include:

- Frequent multicast tree updates: Node mobility and topology changes require frequent updates to the multicast tree.
- Multicast tree maintenance overhead: Maintaining the multicast tree can result in significant control message overhead.

8. Lightweight Adaptive Multicast (LAM) Protocol

LAM is a multicast routing protocol designed for MANETs. Its key features include:

- Adaptive multicast tree construction: LAM dynamically adjusts the multicast tree based on network conditions.
- Lightweight control message overhead: LAM minimizes control message overhead by using a distributed, adaptive approach.

LAM adapts to changing network conditions by:

- Monitoring network topology changes
- Adjusting the multicast tree accordingly

Potential trade-offs between routing efficiency and overhead in highly dynamic environments include:

- Balancing multicast tree optimality with control message overhead
- Managing the frequency of multicast tree updates to minimize overhead while maintaining routing efficiency.

Paper 2

Part A

1. Multipoint Relays (MPRs)

Multipoint Relays (MPRs) are a concept in MANETs where a node selects a subset of its neighbors to retransmit its broadcast messages, reducing redundant transmissions and improving network efficiency.

2. Building blocks of Position-Based Routing (PBR)

1. Location Service: Provides node location information.
2. Position-Based Routing Protocol: Routes packets based on node locations.
3. Location-Aware MAC Protocol: Supports location-aware routing.

3. Drawbacks of Flooding-Generated Broadcast Storm

1. Network congestion
2. Packet collisions
3. Energy inefficiency
4. Scalability issues

4. Applications of MANETs

1. Military communications
2. Emergency response
3. Wireless sensor networks
4. Vehicular ad-hoc networks
5. Personal area networks

5. LBM (Location-Based Multicast) Schemes

1. LBM Scheme 1: Uses a fixed multicast region.
2. LBM Scheme 2: Dynamically adjusts the multicast region based on member locations.

6. Challenging Technologies in Wireless Mesh Networks

1. Scalability and capacity
2. Interference management
3. Quality of Service (QoS) provisioning
4. Security and authentication
5. Node mobility and handover management

Part B

1. Multicast Protocol Supporting Differentiated Service

To support differentiated service, the multicast protocol can use:

1. Priority-based routing: Assign different priorities to different platoons or message types.
2. Quality of Service (QoS) provisioning: Ensure that high-priority messages receive sufficient bandwidth and low latency.
3. Dynamic multicast group management: Allow the commander to dynamically add or remove platoons from the multicast group.
4. Secure multicast protocols: Use secure multicast protocols such as Secure Multicast Routing Protocol (SMRP) or Multicast Security Protocol (MSP).

2. DREAM (Distance Routing Effect Algorithm for Mobility)

DREAM is a routing protocol designed for MANETs. It uses a distance-effect algorithm to predict the future location of mobile nodes, enabling proactive route maintenance.

Key features of DREAM include:

1. Distance-effect algorithm: Predicts the future location of mobile nodes based on their current location, speed, and direction.
2. Proactive route maintenance: Updates routes proactively based on predicted node locations.
3. Adaptive routing: Adapts to changing network conditions and node mobility.

3. GeoGrid and Geocasting for Targeted Advertisements

To track user movement and update ad delivery, GeoGrid and geocasting can use:

1. Location-based services: Obtain user location information from GPS or cellular networks.
2. Grid-based geocasting: Divide the city into a grid and associate each cell with a unique identifier.
3. Dynamic ad delivery: Update ad delivery based on user movement and location.
4. Real-time processing: Process user location information in real-time to ensure timely ad delivery.

4. DSDV Protocol

DSDV (Destination-Sequenced Distance-Vector) is a proactive routing protocol designed for MANETs. It maintains a routing table with the shortest distance to each destination node and uses sequence numbers to ensure routing table consistency.

Key features of DSDV include:

1. Proactive routing: Maintains a routing table with the shortest distance to each destination node.
2. Sequence numbers: Uses sequence numbers to ensure routing table consistency.

3. Periodic updates: Updates routing tables periodically to adapt to changing network conditions.

Part C

Multicasting in a Network without Multicast Capability

Options for multicasting in a network without multicast capability include:

1. Unicast-based multicasting: Send multiple unicast packets to each group member.
2. Application-layer multicasting: Use application-layer protocols to implement multicasting.
3. Overlay networks: Create an overlay network with multicast capability.
4. Tunneling: Use tunneling protocols to encapsulate multicast packets in unicast packets.

6. Traffic Characterization in Wireless Mesh Networks

Traffic in wireless mesh networks is characterized by:

1. Variable traffic patterns: Traffic patterns can change dynamically due to node mobility and changing network conditions.
2. High network congestion: Wireless mesh networks can experience high network congestion due to the shared wireless medium.
3. Quality of Service (QoS) requirements: Wireless mesh networks often require QoS provisioning to support real-time applications.

Compared to typical ad-hoc networks, wireless mesh networks have:

1. Higher network density: Wireless mesh networks have a higher node density, which can lead to increased network congestion.
2. More complex network topology: Wireless mesh networks have a more complex network topology, which can make routing and network management more challenging.

7. Voronoi Diagram-Based Geocasting

Voronoi diagram-based geocasting uses a Voronoi diagram to divide the geographic area into cells, each associated with a unique identifier. Geocasting is performed by forwarding packets to the cells that overlap with the destination region.

Key features of Voronoi diagram-based geocasting include:

1. Voronoi diagram construction: Constructs a Voronoi diagram based on the locations of nodes in the network.
2. Cell-based geocasting: Performs geocasting by forwarding packets to the cells that overlap with the destination region.
3. Adaptive geocasting: Adapts to changing network conditions and node mobility.

8. Routing Protocol for Sensor Networks

To ensure that data from all nodes eventually reaches the base station, the routing protocol can use:

1. Multi-path routing: Use multiple paths to forward data to the base station.
2. Dynamic route maintenance: Update routes dynamically in response to node mobility and changing network conditions.
3. Data aggregation: Aggregate data from multiple nodes to reduce the amount of data transmitted.
4. Energy-efficient routing: Use energy-efficient routing protocols to minimize energy consumption and prolong network lifetime.